

Qindong Gan

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Tokyo, Japan | [frankiegan912.github.io](https://github.com/frankiegan912) | [GitHub](#)

RESEARCH INTERESTS

Multi-agent reinforcement learning; decentralized decision-making; reliable policy evaluation.

EDUCATION

Waseda University

Tokyo, Japan

Master of Engineering in Computer Science and Communications Engineering

09/2025-08/2027 (Expected)

- **GPA: 3.33/4.00 (Waseda scale; A+ in 8 of 12 completed courses)**

University of Cincinnati

Cincinnati, OH, US

Bachelor of Science in Electrical Engineering

09/2020-05/2025

- **GPA: 3.571/4.000**

Joint program with Chongqing University; final academic year in Cincinnati.

Chongqing University

Chongqing, China

Bachelor of Engineering in Electrical Engineering and Automation

09/2020-06/2025

- **GPA: 3.35/4.00**

RESEARCH OUTPUTS

- Q. Gan, Z. Pan, T. D. Ho, J. Wu, and S. Shimamoto, "Causally Verified QoS-Floor Constraints for Decentralized SAGIN Access Association." Manuscript submitted to IEEE CCNC 2027.

RESEARCH EXPERIENCE

Constrained Multi-Agent Learning and Feedback Diagnosis

03/2026-Present

First-author research | Advisor: Shigeru Shimamoto, Waseda University

- Modeled decentralized access association in a space-air-ground network as an interference-coupled constrained Markov game; built the EPyMARL simulation and implemented MAPPO with centralized training and decentralized execution.
- Designed controlled interventions and initialization ablations to test whether Lagrange multipliers conveyed constraint feedback, identifying apparent gains caused by episode-level multiplier resets.
- Reformulated the constraint as a one-sided expected-shortfall budget and compared performance against unconstrained MAPPO, fixed penalties, and heuristic allocation policies.
- In 12-vehicle simulations, increased the mean per-slot minimum rate from 0.011 to 0.192 Mbps, while sum throughput decreased from 252.2 to 222.7 Mbps, relative to unconstrained MAPPO.

ShinsaEvolve

08/2026-Present

Independent research project and technical report | [Code](#) | [Report](#)

- Designed and implemented an LLM-driven recipe-evolution system to test whether candidate-reported training scores reliably select stronger reinforcement learning policies.
- Evaluated trained policies on fresh random seeds using raw environment rewards, with candidate recipe code excluded from the independent audit.
- Ran a 2 x 2 LunarLander study with 12 independent outer-loop runs and 180 candidates. Fixed raw training rewards removed score saturation, but training-to-audit ranking errors persisted.

RGB-Thermal Semantic Segmentation for Autonomous Driving

07/2025-08/2025

Full-time AI Algorithm Intern | Supervised by Jiachen Liang, SERES Group Co., Ltd.

- Adapted DFormerV2's geometry-prior approach from RGB-D segmentation to RGB-thermal semantic segmentation for autonomous driving.
- Collected project data and built, trained, and validated the RGB-thermal segmentation model.

Bio-inspired Mapping and Navigation for Robots

09/2024-05/2025

Senior Capstone Project | University of Cincinnati | Advisor: Ali Minai

- Collected camera images in Webots for visual learning in a hippocampal place-cell navigation model.
- Contributed to EfficientNetV2 model training and coordinated team discussions on model performance and improvements.

PRESENTATIONS

- "Multi-Agent Reinforcement Learning for Intelligent Access Management in Space-Air-Ground Integrated Networks." ASPIRE-CRC-TCA International Workshop, Memorial University, Aug 2026.

HONORS AND AWARDS

- Best Master's Student Presentation Award, ASPIRE-CRC-TCA International Workshop, Memorial University, 2026.
- Dean's List, University of Cincinnati, Fall 2024 and Spring 2025.
- Cooperative Education & Practice - Excellence, University of Cincinnati.
- Second Prize, National College Student Robot Competition.
- Outstanding Student Scholarship (Top 10%); Outstanding Graduate, Chongqing University.

TEACHING EXPERIENCE

Teaching Assistant, University of Cincinnati

09/2024-05/2025

- Supported undergraduate engineering instruction in English through recitations and labs; graded assignments and examinations.

SKILLS

Programming and tools: Python, MATLAB, PyTorch, EPyMARL, NumPy, pandas, scikit-learn, Git, LaTeX.

Methods: Multi-agent and constrained reinforcement learning, Lagrangian methods, simulation, ablation studies, multimodal perception.

Languages: Mandarin Chinese (native), English.